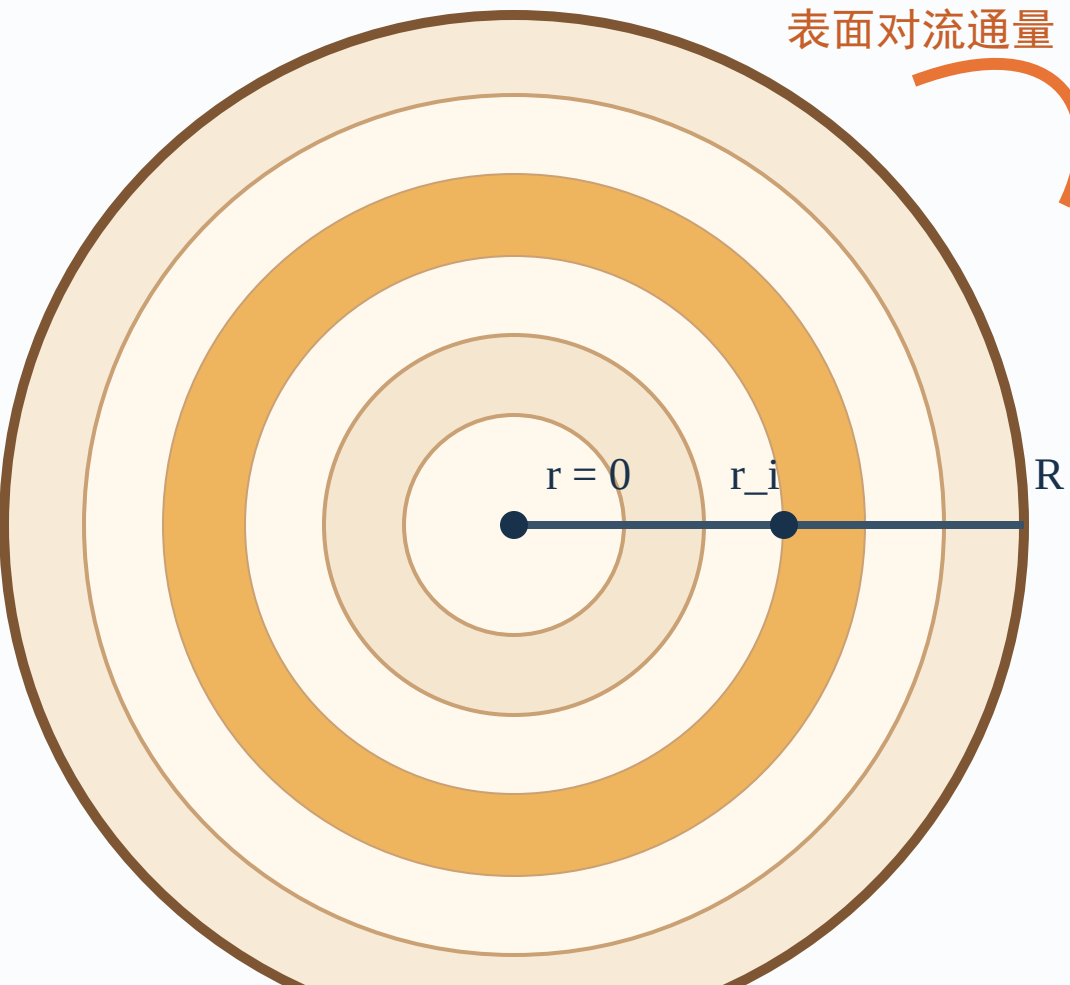


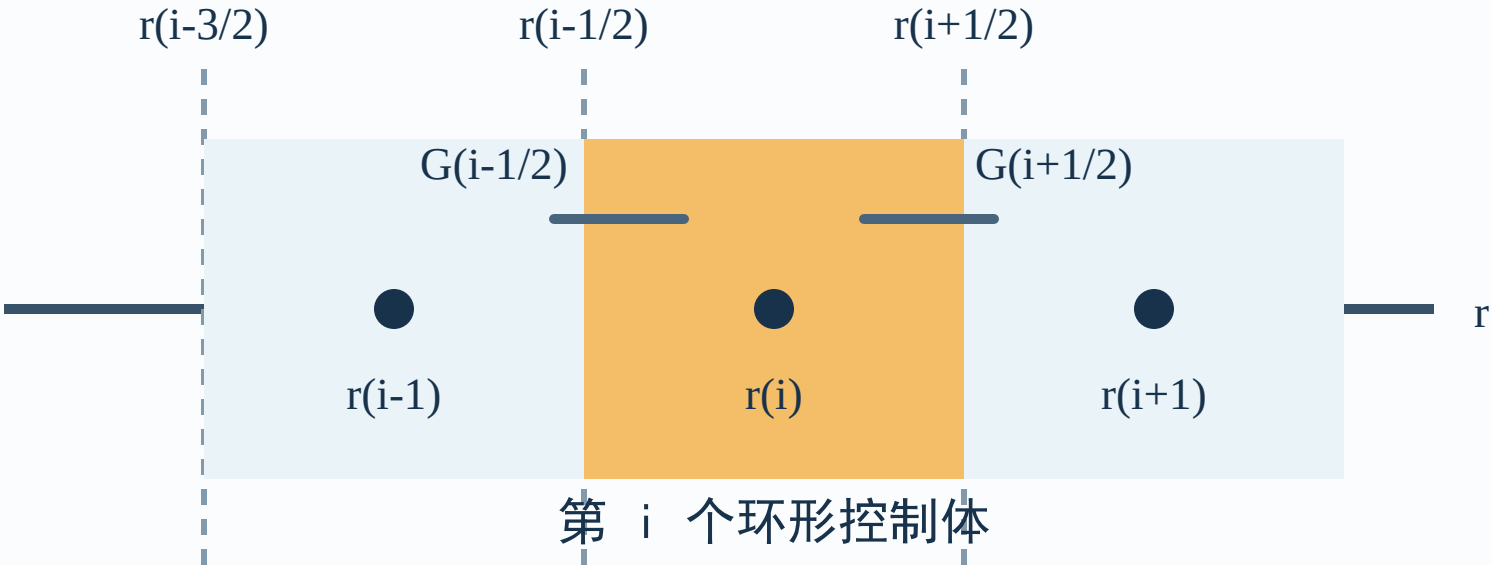
一维径向有限体积离散：环形守恒单元

圆柱横截面与环形控制体



中心控制体内侧面积 $A(-1/2) = 0$
自然满足轴线对称，避免圆柱坐标形式奇点

展开为一维径向网格



对控制体 i 的有符号贡献
 $\Phi_L = G(i-1/2)[u(i-1)-u(i)]$
 $\Phi_R = G(i+1/2)[u(i+1)-u(i)]$
储存项 = $\Phi_L + \Phi_R$; 公共面贡献在全域求和后抵消